



Measurement of Variables

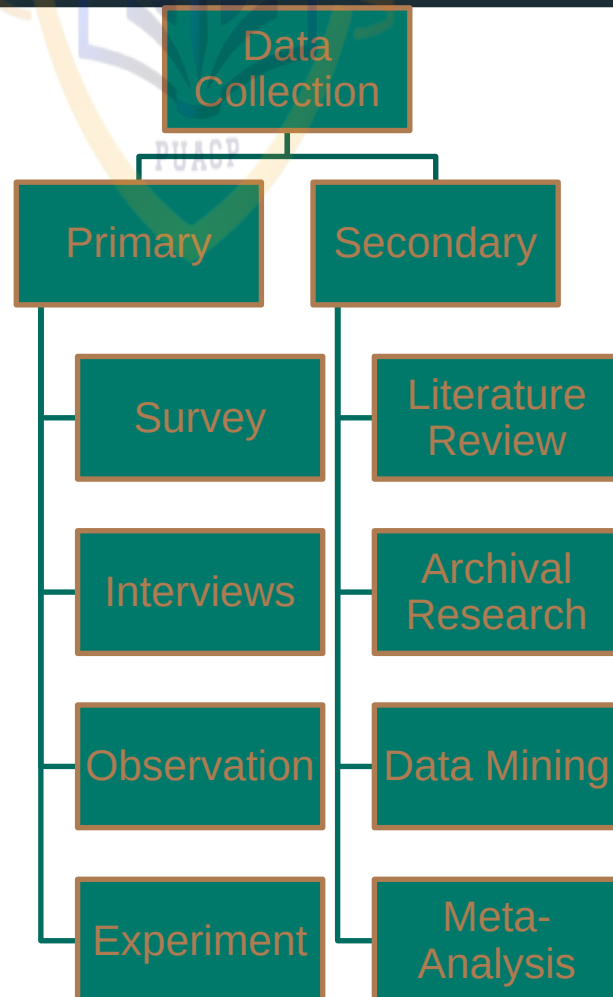
Book: Research Methods for Business: A Skill-Building Approach
Book by Roger Bougie and Uma Sekaran
Chapter: 11 & 12

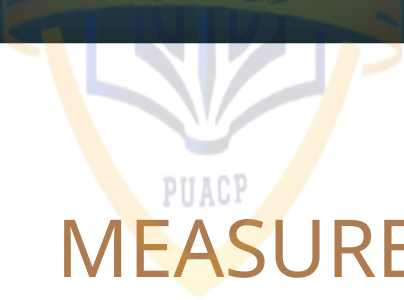


Research

Qualitative

Quantitative





MEASUREMENT

Measurement is the assignment of numbers, codes or other symbols to characteristics (or attributes) of objects according to a prespecified set of rules:

- You cannot measure *objects* (for instance, a company);
- You measure *characteristics or attributes* of objects (for instance, the organizational effectiveness of a company);
- You can measure the length (*the attribute*) of a person (*the object*), the weight of an elephant, the shopping enjoyment of women, the service quality of a restaurant, the conditioning effects of a shampoo, and the taste of a certain brand of yogurt.
- A *judge* is someone who has the necessary knowledge and skills to assess “the quality”.



Class Activity

Identify the **object** and the **attribute**. Give your informed opinion about who would be an adequate **judge**.

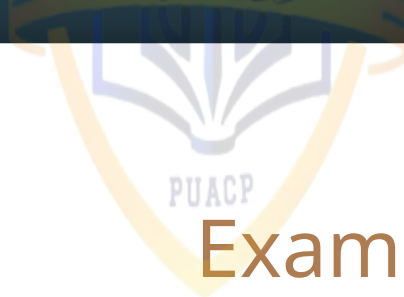
1. **Price consciousness of Fridge buyers**
2. **Self-esteem of school children.**
3. **Stress of household consumers due to inflation**
4. **Marketing performance of automobile companies**
5. **Quality of iPhone**



Operationalization

Reduction of abstract concepts to render them measurable in a tangible way:

- **Conceptual Definition:** Come up with a definition of the construct that you want to measure
- **Define the Dimensions**
- Think about the content of the measure; that is, an instrument (**one or more items or questions**) that actually measures the concept that one wants to measure.
- A **response format** (for instance, a seven-point rating scale with end-points anchored by “strongly disagree” and “strongly agree”) is needed.
- Finally, the **validity and reliability** of the measurement scale has to be assessed.



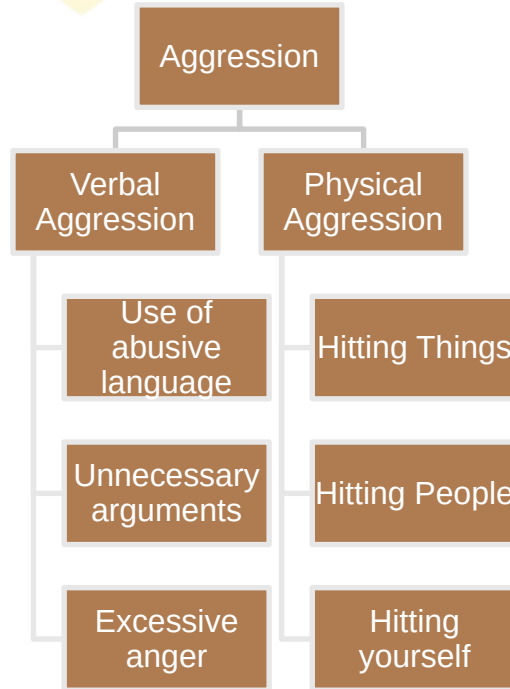
Example

Job Satisfaction is defined as: “one’s positive feelings about job”.

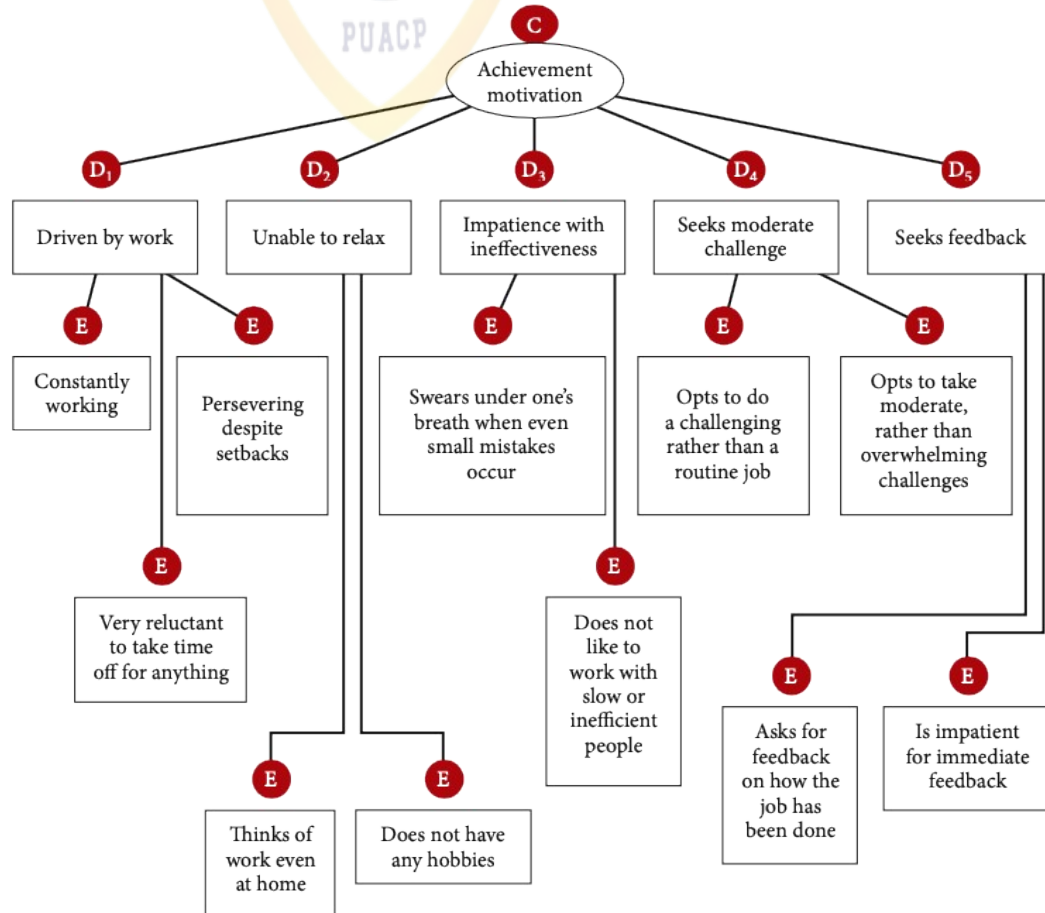
Four items:

1. Overall, I am satisfied with my job.
2. In general, I like my job.
3. In general, I don’t like working here.
4. Overall, I don’t like my job

Example: Operationalizing “Aggression”



Example: "Achievement Motivation"



Questionnaire: Achievement Motivation

Answers to the following questions from respondents would be one way of tapping the level of achievement motivation.

1. To what extent would you say you push yourself to get the job done on time?
2. How difficult do you find it to continue to do your work in the face of initial failure or discouraging results?
3. How often do you neglect personal matters because you are preoccupied with your job?
4. How frequently do you think of your work when you are at home?
5. To what extent do you engage yourself in hobbies?
6. How disappointed would you feel if you did not reach the goals you had set for yourself?
7. How much do you concentrate on achieving your goals?
8. How annoyed do you get when you make mistakes?
9. To what extent would you prefer to work with a friendly but incompetent colleague, rather than a difficult but competent one?
10. To what extent would you prefer to work by yourself rather than with others?
11. To what extent would you prefer a job that is difficult but challenging, to one that is easy and routine?
12. To what extent would you prefer to take on extremely difficult assignments rather than moderately challenging ones?
13. During the past three months, how often have you sought feedback from your superiors on how well you are performing your job?
14. How often have you tried to obtain feedback on your performance from your coworkers during the past three months?
15. How often during the past three months have you checked with your subordinates that what you are doing is not getting in the way of their efficient performance?
16. To what extent would it frustrate you if people did not give you feedback on how you are progressing?



Class Activity

Operationalize “Happiness”





"SCALING"

*"Let's assign numbers or
symbols to responses"*





Scales

A scale is a tool or mechanism by which individuals are distinguished as to how they differ from one another on the variables of interest to our study



Types of Scale

1. A **nominal scale** is one that allows the researcher to assign subjects to certain categories or groups.

Example: with respect to the variable of gender, respondents can be grouped into two categories – male and female. These two groups can be assigned code numbers 1 and 2.

2. An **ordinal scale** not only categorizes the variables in such a way as to denote differences among the various categories, it also rank-orders the categories in some meaningful way.

Example: the first-ranked job characteristic might be only marginally preferred over the second-ranked characteristic, whereas the characteristic that is ranked third might be preferred in a much larger degree than the one ranked fourth.



Types of Scale

3. In an **interval scale**, numerically equal distances on the scale represent equal values in the characteristics being measured.

Example: Strongly Agree to Strongly Disagree

4. **Ratio Scale**: not only measures the magnitude of the differences between points on the scale but also taps the proportions in the differences.

Example: Weighing scale has an absolute (and not arbitrary) zero origin calibrated on it, which allows us to calculate the ratio of the weights of two individuals.

- How many other organizations did you work for before joining this system?
- Please indicate the number of children you have?
- How many hours do you work at a job?



Scales

Ranking

“Several response categories and are used to elicit responses with regard to the object, event, or person studied”

Rating

make comparisons between or among objects, events, or persons and elicit the preferred choices and ranking among them



Rating Scales



Dichotomous Scale

- Scale with yes or no answer:

EXAMPLE

Do you own a car?

Yes

No



Category Scale

- Uses multiple items to elicit a single response:

EXAMPLE

Where in London do you reside?

- ☐ East London
- ☐ South London
- ☐ West London
- ☐ North London
- ☐ Outskirts



Semantic Differential Scale

- Several bipolar attributes are identified at the extremes of the scale

EXAMPLE

Responsive	— — — — —	Unresponsive
Beautiful	— — — — —	Ugly
Courageous	— — — — —	Timid



Numerical Scale

- Similar to the semantic differential scale, with the difference that numbers on a five-point or seven-point scale are provided

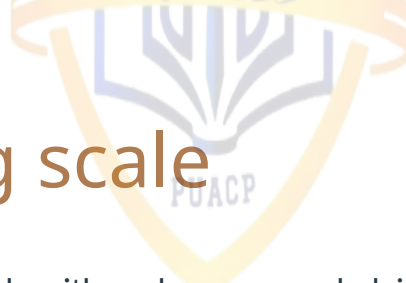
EXAMPLE

How pleased are you with your new estate agent?

Extremely Pleased

7 6 5 4 3 2 1

Extremely Displeased



Itemized rating scale

- A five-point or seven-point scale with anchors, as needed, is provided for each item and the respondent states the appropriate number on the side of each item

Respond to each item using the scale below, and indicate your response number on the line by each item.

3				
1	2	Neither Unlikely	4	5
Very Unlikely	Unlikely	Nor Likely	Likely	Very Likely
1	I will be changing my job within the next 12 months.			—
2	I will take on new assignments in the near future.			—
3	It is possible that I will be out of this organization within the next 12 months.			—



Likert Scale

- The Likert scale is designed to examine how strongly subjects agree or disagree with statements on a five-point scale with the following anchors:

Strongly Disagree	Disagree	Neither Agree nor Disagree	Agree	Strongly Agree
1	2	3	4	5

EXAMPLE

Using the preceding Likert scale, state the extent to which you agree with each of the following statements:

My work is very interesting	1	2	3	4	5
I am not engrossed in my work all day	1	2	3	4	5
Life without my work would be dull	1	2	3	4	5



Fixed or constant sum scale

- The respondents are here asked to distribute a given number of points across various items as per the example below:

EXAMPLE

In choosing a toilet soap, indicate the importance you attach to each of the following five aspects by allotting points for each to total 100 in all.

Fragrance	—
Color	—
Shape	—
Size	—
Texture of lather	—
Total points	100



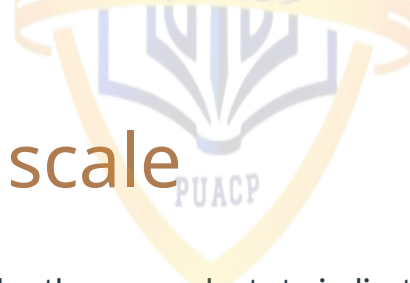
Stapel scale

- This scale simultaneously measures both the direction and intensity of the attitude toward the items under study. The characteristic of interest to the study is placed at the center with a numerical scale ranging, say, from +3 to -3, on either side of the item:

EXAMPLE

State how you would rate your supervisor's abilities with respect to each of the characteristics mentioned below, by circling the appropriate number.

+3	+3	+3
+2	+2	+2
+1	+1	+1
Adopting modern technology	Product innovation	Interpersonal skills
-1	-1	-1
-2	-2	-2
-3	-3	-3



Graphic rating scale

- A graphical representation helps the respondents to indicate on this scale their answers to a particular question by placing a mark at the appropriate point on the line, as in the following example.

EXAMPLE

On a scale of 1 to 10, how would
you rate your supervisor?

– 10 Excellent
–
–
–
–
– 5 Adequate
–
–
–
– 1 Very bad



Consensus scale

Scales can also be developed by consensus, where a panel of judges selects certain items, which in its view measure the relevant concept. The items are chosen particularly based on their pertinence or relevance to the concept.



RANKING SCALE



Paired Comparison

The paired comparison scale is used when, among a small number of objects, respondents are asked to choose between two objects at a time. This helps to assess preferences.

Example: during the paired comparisons, respondents consistently show a preference for product one over products two, three, and four, the manager can reliably understand which product line demands his utmost attention.



Forced Choice

The forced choice enables respondents to rank objects relative to one another, among the alternatives provided. This is easier for the respondents, particularly if the number of choices to be ranked is limited in number.

EXAMPLE

Rank the following magazines that you would like to subscribe to in the order of preference, assigning 1 to the most preferred choice and 5 to the least preferred.

<i>Fortune</i>	—
<i>Playboy</i>	—
<i>Time</i>	—
<i>People</i>	—
<i>Prevention</i>	—



Comparative scale

- The comparative scale provides a benchmark or a point of reference to assess attitudes toward the current object, event, or situation under study

EXAMPLE

In a volatile financial environment, compared to stocks, how wise or useful is it to invest in Treasury bonds?
Please circle the appropriate response.

More useful		About the same		Less useful
1	2	3	4	5



Class Activity

Develop and name the type of measuring instrument you would use to tap the following:

- a. Which brands of toothpaste are consumed by how many individuals?
- b. Among the three types of exams – multiple choice, essay type, and a mix of both – which is the one preferred most by students?
- c. To what extent do individuals agree with your definition of accounting principles?
- d. How much people like an existing organizational policy.
- e. The age of employees in an organization.
- f. The number of employees in each of the 20 departments of a company.